MD SHOAIB MAHMUD

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RESEARCH INTERESTS

Operations research and mixed-integer programming; prescriptive analytics for infrastructure resilience and disaster response; graph neural networks for network vulnerability and retrofit prioritization; humanitarian and supply chain logistics; robust and stochastic optimization under uncertainty.

EDUCATION

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh *Feb 2020 – Apr 2024*

B.Sc. in Industrial and Production Engineering **CGPA: 3.77 / 4.00**

Class rank: 4th out of 51 graduates. Dean's List of Honor (2020–2023). University Merit Scholarship (2021–2023).

Selected coursework: Operations Research, Numerical Analysis, Probability and Statistics, Supply Chain Management, Operations Management, Quality Management.

Undergraduate Thesis: *Reverse Logistics Network Design for the Semiconductor Supply Chain* (Mar 2023 – Apr 2024)

- Formulated a mixed-integer linear program (MILP) integrating forward and reverse flows to mitigate semiconductor chip shortages, implemented and solved in IBM ILOG CPLEX.
- Designed a quick-response recovery network to test and redeploy functional chips, lowering reliance on new facilities and shortening lead times through cost-efficient routing.

PUBLICATIONS

Published Journal Articles

[1] Zaman, T. A. B., **Mahmud, M. S.**, Gupta, H. S., Harati, M., & van de Lindt, J. W. (2026). Quantifying the cost of delay in floodplain property buyouts. *Sustainability, 18*(6), 2675. https://doi.org/10.3390/su18062675

In Conference Proceedings

[1] **Mahmud, M. S.**, Saad, S. I., & Miazi, M. H. (2025). Data-driven application of the newsvendor model for retail inventory optimization using forecast-error demand estimation. *Proceedings of the 8th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management*, Dhaka, Bangladesh. https://doi.org/10.46254/BA08.20250372

[2] Rubaet, M., **Mahmud, M. S.**, Mukit, M. U. A., Mithu, M. S. H., Islam, M. A., & Sakib, M. I. A. (2025). The systematic design and manufacturing of 'Mr. Money Manager': An advanced money management device. *Proceedings of the 8th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management*, Dhaka, Bangladesh. https://doi.org/10.46254/BA08.20250054

[3] Debnath, A., & **Mahmud, M. S.** (2025). Dynamic staffing optimization for institutional dining services: An Erlang C queueing approach. *Proceedings of the 8th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management*, Dhaka, Bangladesh. https://doi.org/10.46254/BA08.20250081

[4] Alam, S. T., Sagor, E., Ahmed, T., Haque, T., **Mahmud, M. S.**, Ibrahim, S., Shahjahan, O., & Rubaet, M. (2023). A comparative analysis of assignment problem. In *Big Data Innovation for Sustainable Cognitive Computing* (pp. 125–142). Springer Nature. https://doi.org/10.1007/978-3-031-28324-6_11

[5] Alam, S. T., Sagor, E., Ahmed, T., Haque, T., **Mahmud, M. S.**, Ibrahim, S., & Rubaet, M. (2022). Assessment of assignment problem using the Hungarian method. *Proceedings of the 1st Australian Conference on Industrial Engineering and Operations Management*, Sydney, Australia.

[6] Alam, S. T., Sagor, E., Ahmed, T., Haque, T., & **Mahmud, M. S.** (2022). A comparative analysis of balanced transportation problems. *Proceedings of the International Conference on Mechanical Engineering and Applied Sciences (ICMEAS 2022)*, MIST, Dhaka, Bangladesh.

Journal Manuscripts Under Review

[1] **Mahmud, M. S.**, Ikbal, M. N., Gupta, H. S., Sanderson, D., & Biswas, S. (2025). Graph neural networks for seismic vulnerability assessment and retrofit prioritization of transportation networks. *Reliability Engineering & System Safety*. (Under review, JRESS-D-25-05531)

[2] Gupta, H. S., **Mahmud, M. S.**, & Gupta, A. (2025). From calibrated risk to actionable response: A prescriptive analytics pipeline for urban flash floods. *Resilient Cities and Structures*. (Under review, RCNS-D-25-00275)

[3] Gupta, H. S., Saad, S. I., **Mahmud, M. S.**, & Biswas, S. (2025). Equity-aware ambulance dispatch after earthquakes. *Safety Science*. (Under review, SAFETY-D-25-03429)

[4] Iqbal, M. N., **Mahmud, M. S.**, Gupta, H. S., Duarte, J. A., Hossain, N. U. I., & Jaradat, R. (2026). *Equity-aware interpretable policy rules for personalized hurricane risk communication*. *International Journal of Disaster Risk Reduction*. (Under review, IJDRR-D-26-00493)

[5] Alam, S. T., Ahmed, T., Haque, T., **Mahmud, M. S.**, & Kaisar, T. I. (2024). Profit-oriented pricing combinations using RSM-ANN. *Discover Applied Sciences*. (Under review)

[6] Saad, S. I., **Mahmud, M. S.**, Tani, T. I., Ishrak, A. M., Mumtaj, S., & Tinni, T. I. (2025). Reducing delivery time in electric VRPs through adaptive partial recharging. *Proceedings of the 4th International Conference on Innovations in Data Analytics*, Kolkata, India. (Accepted)

Manuscripts in Preparation

[1] **Mahmud, M. S.**, Sen Gupta, H., & Wang, W. (2026). Post-tornado power grid restoration: A two-stage robust optimization model with ML-informed vulnerability. Target: *Reliability Engineering & System Safety*.

[2] **Mahmud, M. S.**, Haque, T., & Ahmed, T. (2025). Reverse logistics network design for the semiconductor supply chain: An optimization approach. Target: *Brazilian Journal of Operations and Production Management*.

RESEARCH EXPERIENCE

Research Associate – RISE Lab (Colorado State University Pueblo) *2024 –Present*

Supervised by Dr. Himadri Sen Gupta

- Developing graph neural network models for seismic vulnerability assessment and retrofit prioritization across large-scale transportation networks, co-led methodology design and computational experiments.
- Contributing to a prescriptive analytics pipeline that converts calibrated flood risk forecasts into actionable urban response plans, including scenario generation and decision modeling.
- Building a two-stage robust optimization framework with machine-learning-informed vulnerability for post-tornado power grid restoration.

TEACHING EXPERIENCE

Lecturer, Department of Industrial and Production Engineering, MIST *Nov 2024 – Present*

- Taught five theory courses: IPE 101-Introduction to IPE, IPE 107-Engineering Economy, MATH 215-Numerical Analysis, IPE 31- Ergonomics and Safety Management, and IPE 419-Modeling and Simulation.
- Lead four sessional courses spanning Engineering Graphics and CAD, Probability and Statistics, Operations Research, and Ergonomics and Safety Management.
- Supervise undergraduate thesis students and mentor capstone projects that apply optimization, simulation, and statistical methods to industry problems.
- Modernized course content by integrating CPLEX, Arena, and Python-based analytics tools into the IPE curriculum.

ACADEMIC SERVICE AND PROFESSIONAL EXPERIENCE

Editor, Technical Paper Volume 3, Department of IPE, MIST *Jun 2025 – Oct 2025*

- Led peer review and editorial coordination for the departmental research volume featuring student and faculty work, ensuring methodological rigor and clarity.

Director of Program, IEOM MIST Student Chapter *2023 – 2024*

- Organized technical workshops on supply chain management and MATLAB for the student body, training over 100 participants.

Industrial Trainee, British American Tobacco Bangladesh *Feb 2023*

- Delivered the 'Manual Management' and 'Project Apollo' analytics initiatives, building Power BI dashboards over large operational datasets; recognized as a top-performing trainee of the cohort.

TECHNICAL SKILLS

Programming: Python, C, C++

Optimization and Modeling: IBM ILOG CPLEX, MILP, robust and stochastic optimization, network optimization

Machine Learning: Graph neural networks, regression and classification pipelines, scikit-learn

Simulation and Statistics: Arena, Minitab, SPSS, Power BI

Engineering Design: SOLIDWORKS, CATIA

Scientific Writing: LaTeX, Microsoft Office, reference management

HONORS AND AWARDS

- Dean's List of Honor, MIST (2020–2023) — awarded for sustained CGPA above 3.75.
- University Merit Scholarship, MIST (2021–2023) — awarded for top three class positions across consecutive semesters.
- Top-Performing Trainee, British American Tobacco Bangladesh Industrial Attachment Program (2023).